Microcontroller

File: microcontroller.kicad_sch

Sensors

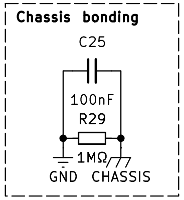
File: sensors.kicad_sch

RF

File: rf.kicad_sch

Power

File: power.kicad_sch



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Flight Computer

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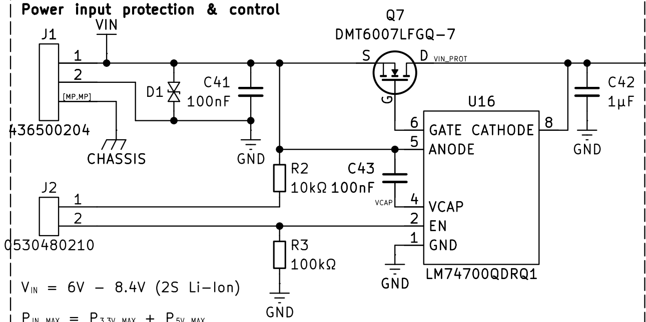
Power input protection & control

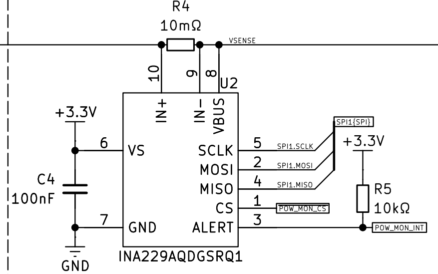


V_{IN} = 6V – 8.4V (2S Li-Ion)

$P_{IN\ MAX} = P_{3.3V\ MAX} + P_{5V\ MAX} + P_{SENSE\ MAX} + P_{PYRO\ CH\ MAX}$ =
11.65W + 16.67W + 0.34W + 6.22W = 34.88W

$I_{IN\ MAX} = P_{IN\ MAX} / V_{IN\ MIN}$ =
34.77W / 6V = 5.81A

Power monitoring

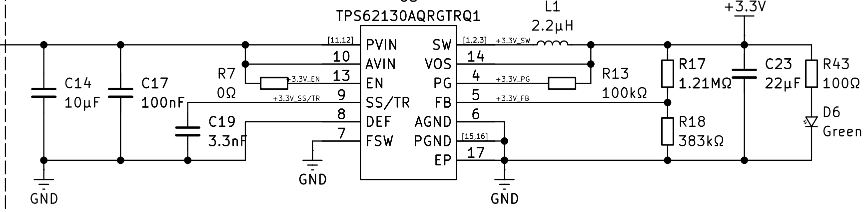


$V_{SENSE\ MAX}$ = 163.84mV

$R_{SHUNT\ MAX} = V_{sense\ max} / I_{max}$ =
163.84mV / (34.54W / 6V) = 28.46mΩ = 10mΩ

$P_{max} = R_{shunt} \times I_{max}^2$ =
10mΩ × 5.79A² = 0.34W

3.3V DC/DC Regulator

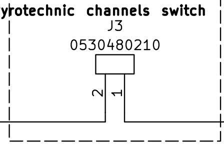


$V_{OUT} = ((R_{TOP} / R_{BOTTOM}) + 1) \times 0.8V$ =
((1.21MΩ / 383kΩ) +1) × 0.8V =
3.33V

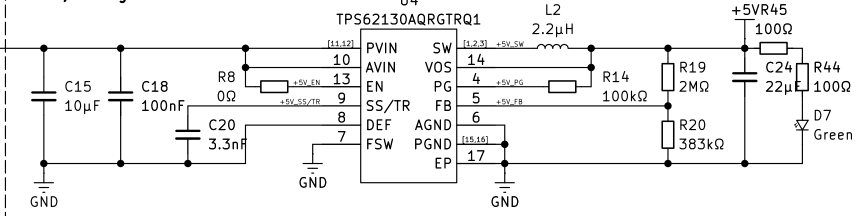
$I_{LED} = (V_{IN} - V_F) / R$ =
3.3V – 2.2V / 100Ω = 11mA

$P_{IN\ MAX} = (P_{3.3V} / \eta_{3.3V\ @\ 3A})$ =
((3.3V × 3A) / 0.85) =
11.65W

Pyrotechnic channels switch



5V DC/DC regulator

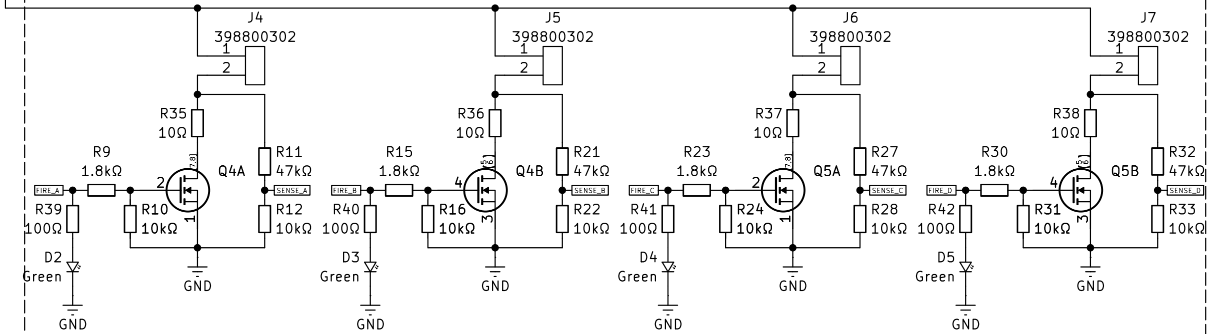


$V_{OUT} = ((R_{TOP} / R_{BOTTOM}) + 1) \times 0.8V$ =
((2MΩ / 383kΩ) +1) × 0.8V =
4.98V

$I_{LED} = (V_{IN} - V_F) / R$ =
5V – 2.2V / 200Ω = 14mA

$P_{IN\ MAX} = P_{5V} / \eta_{5V\ @\ 3A}$ =
((5V × 3A) / 0.9) =
16.67W

Pyrotechnic channel A,B,C,D



R_{LIMIT} = 10Ω
t = 100ms

$I_{MAX\ CH} = V_{MAX} / R_{MIN} + R_{LIMIT}$ =
8.4V / (1.2Ω + 10Ω) = 0.74A

$I_{MIN\ CH} = V_{MIN} / R_{MAX} + R_{LIMIT}$ =
6V / (1.8Ω + 10Ω) = 0.51A

$P_{LIMIT\ MAX} = R_{LIMIT} \times I_{MAX}^2$ =
10Ω × 0.74A² = 5.48W

$P_{PYRO\ CH\ MAX} = V_{MAX} \times I_{MAX}$ =
8.4V × 0.74A = 6.22W

$I_{SENSE\ MAX} = V_{MAX} / R_{TOT}$ =
8.4V / (47kΩ + 10kΩ) = 0.15mA

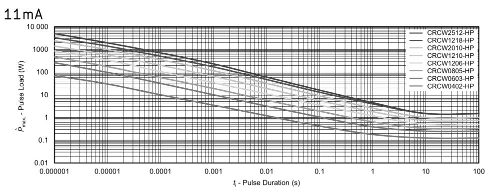
$I_{SENSE\ MIN} = V_{MIN} / R_{TOT}$ =
6V / (47kΩ + 10kΩ) = 0.10mA

$V_{SENSE\ MAX} = V_{MAX} \times (R_{BOTTOM} / (R_{TOP} + R_{BOTTOM}))$ =
8.4V × (10kΩ / (47kΩ+ 10kΩ)) = 1.47V

$V_{SENSE\ MIN} = V_{MIN} \times (R_{BOTTOM} / (R_{TOP} + R_{BOTTOM}))$ =
6V × (10kΩ / (47kΩ+ 10kΩ)) = 1.05V

$I_{LED} = (V_{IN} - V_F) / R$ =
3.3V – 2.2V / 100Ω = 11mA

Single Pulse



Maximum pulse load, single pulse; applicable if $\bar{P} \to 0$ and n < 1000 and $\hat{U} \le \hat{U}_{max}$;
for permissible resistance change equivalent to 8000 h operation

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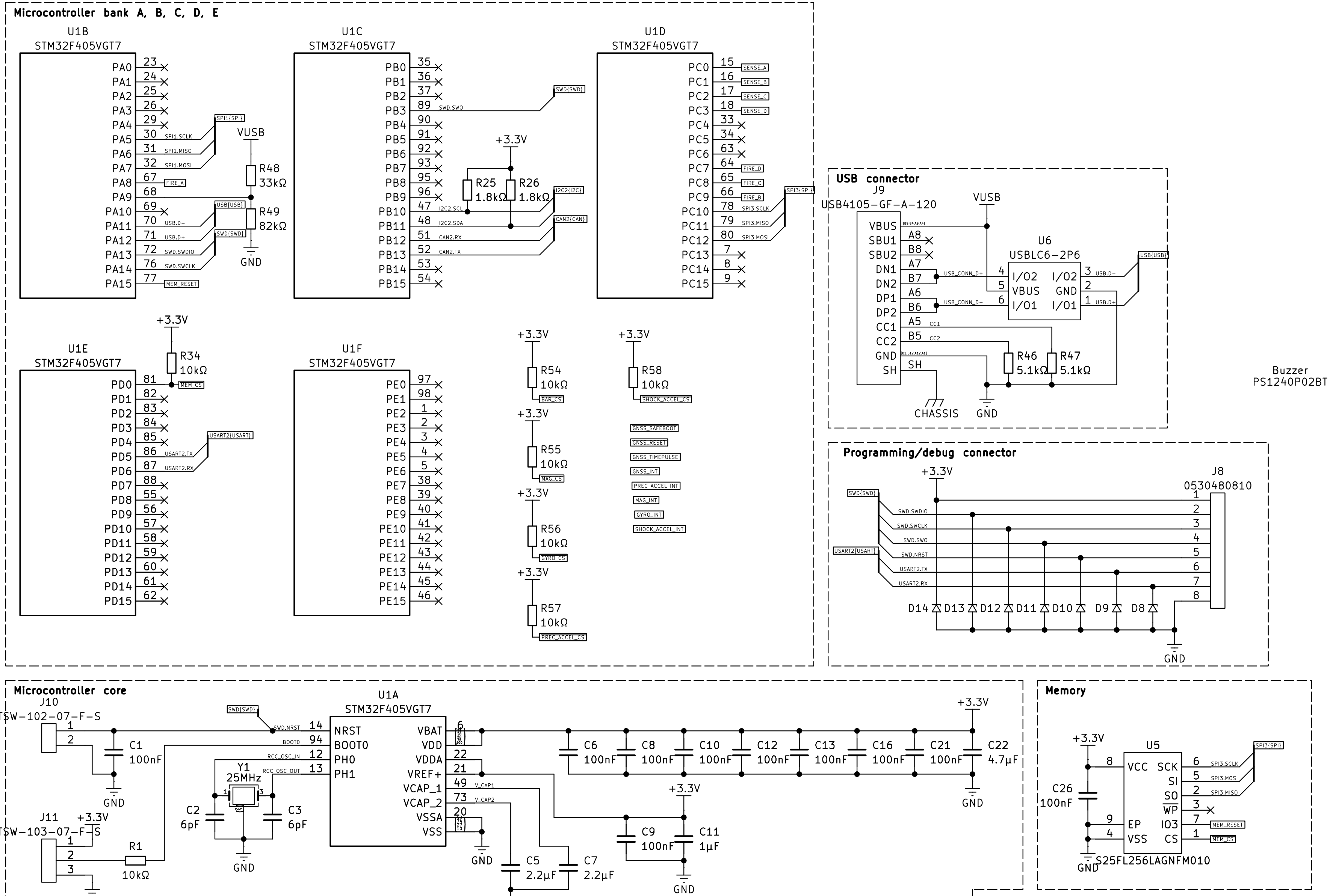

F = 25MHz

ESR = 40Ω

C_L = 8pF

C_0 = 7pF

C_S = 5pF

g_m = 5 mA/V

$C_{tot} = C_L + (C_S / 2)$

Table 4. Safety factor (S_f) for STM32 and STM8 oscillators(1)

Safety factor (S_f)	Assurance level	
	HSE	LSE
$S_f \geq 5$	Safe	Very safe
$3 \leq S_f < 5$	Not safe	Safe
$S_f < 3$		Not safe

1. Safe and very safe oscillations are shown in green, unsafe oscillation in yellow.

$C_{L1/2} = 2 \times (C_L - C_S) = 6pF$

$DL = (ESR \times (\pi \times F \times C_{tot})^2 \times (V_{pp})^2) / 2 = 97.07\mu W$

$g_{m\ critical} = 4 \times (ESR) \times (2 \times \pi \times F)^2 \times (C_0 + C_L)^2 = 0.89$ mA/V

$g_{m\ margin} = g_{m\ margin} / g_{m\ critical} = 5.63$

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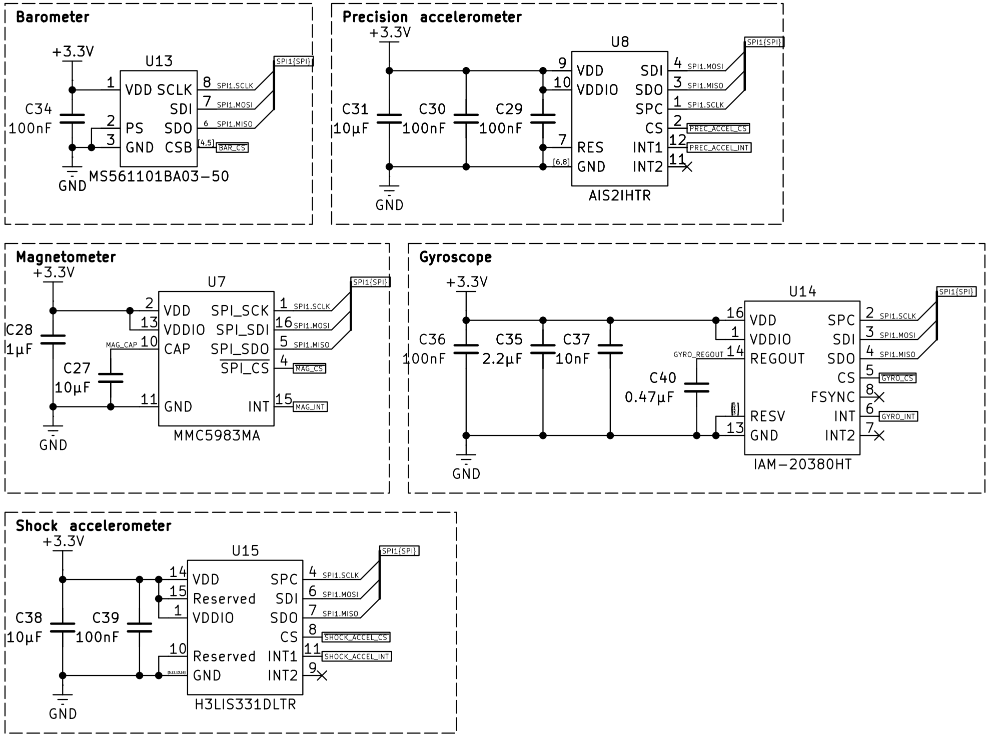

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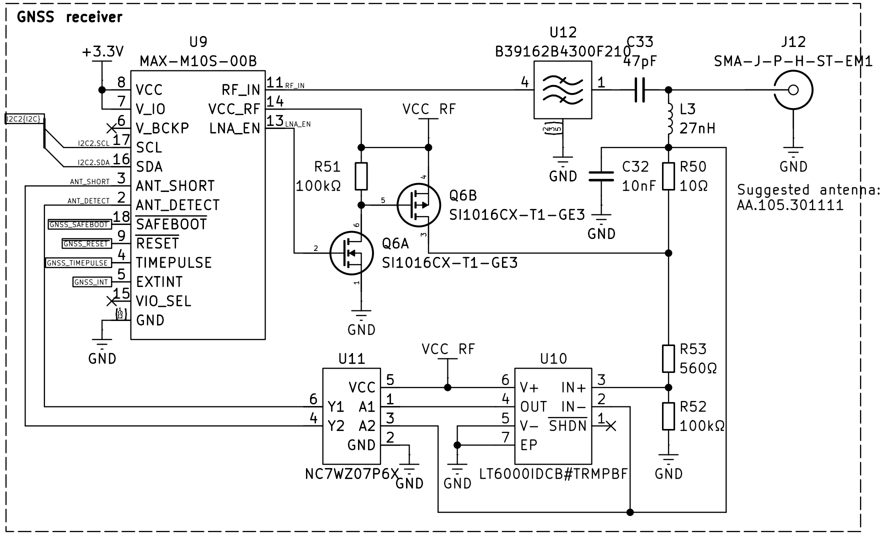

Radio
CC1200
CC1201

Organization:	Lapin Rocketry
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